The eigenvalue decay of the limiting kernel of a sparse Mertens matrix

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Abstract

Order the squarefree integers as a rooted tree in which the parent of $d > 1$ is $d/P^+(d)$, where $P^+(d)$ is the largest prime factor of d. On this tree consider the symmetric kernel K whose entry at an ancestor j and a descendant k is $\mu(k/j)\delta_k$, where $\delta_k = (6/\pi^2)k^{-1}\prod_{p\le P^+(k)} p/(p+1)$. It arises as the limit of normalized Gram matrices of inverses of sparse $(0,1)$ matrices whose bordered versions have determinant equal to the Mertens function. An earlier release by the author identified each fixed eigenvalue of K as a limit of finite singular values; here we determine how the eigenvalues decay. We prove that K is positive and compact with

$$\lambda_r(K) \sim \frac{6/\pi^2}{r\log r} \qquad (r\to\infty),$$

equivalently that the number of eigenvalues exceeding s is asymptotic to $(6/\pi^2)/(s\log(1/s))$. The same law holds for a rank-one deflation of K whose eigenvalues determine, index by index, the limits of the rescaled lower singular values of the bordered matrices, apart from the smallest one. The proof removes the Möbius signs by a diagonal unitary, realizes the kernel as a Gram matrix of nested cylinder sets for a random set of primes, and shows that the eigenvalue count is asymptotic to the number of cylinders of mass above the threshold. That number is evaluated with Mertens' theorem and the Dickman function; the constant comes from the identity $\int_0^\infty \rho = e^\gamma$. No convergence rate is obtained, and no estimate for the Mertens function follows.

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1 Introduction

All logarithms are natural. Write $\mathcal{S}$ for the positive squarefree integers, μ for the Möbius function, and $P^+(d)$ for the largest prime factor of d, with $P^+(1) = 1$. Put

$$c_0 = \frac{6}{\pi^2}, \qquad \nu_d = \frac{1}{d} \prod_{p \le P^+(d)} \frac{p}{p+1}, \qquad \delta_d = c_0 \nu_d \quad (d \in \mathcal{S}).$$

Thus $\nu_1 = 1$. The primes in each product are unrestricted by divisibility.

The squarefree integers form a rooted tree: the root is 1, and the parent of $d > 1$ is $d/P^+(d)$. Write $j \preceq k$ when j is an ancestor of k, including equality. Equivalently, $j \mid k$ and every prime divisor of k/j exceeds $P^+(j)$. Listing the prime factors of k in increasing order, the ancestors of k are exactly the products of its initial segments.

Define a real symmetric matrix on $\mathbb{N} \times \mathbb{N}$ by

$$K_{jk} = \begin{cases} \mu(k/j)\delta_k, & j,k \in \mathcal{S},\ j \preceq k, \\ \mu(j/k)\delta_j, & j,k \in \mathcal{S},\ k \preceq j, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The two applicable cases agree on the diagonal. Nonsquarefree coordinates are zero. We shall show that this matrix defines a positive compact operator on $\ell^2(\mathbb{N})$.

For a positive compact operator A, let

$$\mathsf{n}(A;t) = \#\{r : \lambda_r(A) > t\}, \qquad t > 0,$$

where positive eigenvalues are ordered decreasingly, with multiplicity. All counting functions in the operator arguments use this strict convention. The subject of this paper is the rate at which the eigenvalues of K tend to zero.

Theorem 1.1. *The operator K in (1) is positive and compact. As $s \downarrow 0$,*

$$\mathsf{n}(K; s) \sim \frac{c_0}{s \log(1/s)}. \tag{2}$$

Equivalently, as $r \to \infty$,

$$\lambda_r(K) \sim \frac{c_0}{r \log r}. \tag{3}$$

Consequently,

$$\sum_{r \le R} \lambda_r(K) \sim c_0 \log\log R.$$

The same three asymptotics hold for the positive compact operator

$$L = K - \frac{(Ke_1)(Ke_1)^*}{c_0}. \tag{4}$$

Two immediate consequences are that K and L lie in the Schatten class $\mathcal{S}_p$ exactly when $p > 1$ (Corollary 5.1), and that K is not trace class even though K minus its diagonal is (Proposition 5.2).

1.1 Where the kernel comes from

The kernel K appeared in the author's earlier study [12] of a sparse matrix, introduced in [11], whose determinant is the Mertens function $M(n) = \sum_{m \le n} \mu(m)$. Let A_n be the $n \times n$ matrix with ones on the diagonal and an additional one in position $(i, i/P^+(i))$ for each squarefree i with $1 < i \le n$. Its inverse $C_n = A_n^{-1}$ has entries $\mu(i/j)$ when $j \preceq i$ (Section 6). Replacing the first row of A_n by ones gives a matrix B_n with $\det B_n = M(n)$. The following facts are proved in [12].

- [12, Theorem 4.3]: the matrices $C_n^* C_n / n$, extended by zero, converge to K in Hilbert–Schmidt norm. Hence, for every fixed r, $s_r(C_n)/\sqrt{n} \to \sqrt{\lambda_r(K)}$.
- [12, Theorem 4.6]: for every fixed $r \ge 1$, $\sqrt{n}\, \sigma_{n-r}(B_n) \to \lambda_r(L)^{-1/2}$, with L as in (4).

Both statements fix the index r and let $n \to \infty$. They identify each limit but do not describe how the limits behave as r grows. The earlier analysis also gives little quantitative information about the limiting operator itself: Hilbert–Schmidt membership gives $\sum_r \lambda_r(K)^2 < \infty$, but not the order of $\lambda_r(K)$. Theorem 1.1 answers this question for the limiting operators. It does not, by itself, give information about $s_r(C_n)$ when r and n grow together; Section 6 records what an operator-norm error estimate would add.

A first guess comes from the diagonal. The diagonal entries of K are δ_d, and Mertens' product theorem gives $\delta_d \approx e^{-\gamma}/(d \log P^+(d))$ when $P^+(d)$ is large. The off-diagonal part is not small in the sense needed, however. It is trace class (Proposition 5.2), which controls partial sums of eigenvalues up to a bounded error. The eigenvalues themselves, however, are of order $1/(r \log r)$, and a trace-class operator may have singular values that are not $o(1/(r \log r))$. Trace-class membership alone therefore does not transfer an asymptotic for individual eigenvalues from the diagonal to K. The proof uses the structure of K instead.

1.2 Outline of the proof

The argument has four steps.

Removing the signs. On comparable squarefree indices $\mu(k/j) = \mu(j)\mu(k)$, so conjugation by the diagonal unitary with entries $\mu(d)$ replaces K by the unsigned kernel with entries $\delta_{\max(j,k)}$ on comparable pairs. The spectrum is unchanged.

A cylinder representation. Choose a random set of primes, including each prime p independently with probability $1/(p+1)$. For $d \in \mathcal{S}$ let A_d be the event that the selected primes up to $P^+(d)$ are exactly the prime divisors of d. Then A_d has probability ν_d; the events are nested along the tree and disjoint for incomparable indices. The unsigned kernel is therefore c_0 times the Gram matrix of the indicators $\mathbf{1}_{A_d}$ (Section 2).

From masses to eigenvalues. Section 3 treats any tree of nested sets whose children are disjoint and carry at most a fixed fraction $\theta < 1$ of their parent's mass. The operator T, the sum over vertices v of the rank-one maps $f \mapsto \langle f, \mathbf{1}_{A_v} \rangle \, \mathbf{1}_{A_v}$, is compact, and splitting each subtree into its root and its children expresses T as a rank-one operator plus an orthogonal sum of rescaled copies of the subtree operators. Rank-one interlacing, applied recursively, traps the eigenvalue count of T between the number of leaves and an inflated number of vertices of the finite tree $\{v : \nu(A_v) > t\}$. When the child-to-parent mass ratios tend to zero outside finite sets and the mass count is regularly varying, both bounds are asymptotic to the mass count (Theorem 3.4).

Counting the masses. It remains to count d with $\nu_d > t$ (Section 4). Grouping d by its largest prime factor reduces this to counts of squarefree smooth numbers, which we evaluate with the Dickman function and Mertens' theorems. The answer is $D(t) \sim 1/(t \log(1/t))$. Its constant is the product of the squarefree density c_0, the limit $\zeta(2)e^{-\gamma}$ of $m(p) \log p$, where $m(p) = \prod_{q \le p} q/(q+1)$, and the classical value $\int_0^\infty \rho(v)\, dv = e^\gamma$; since $c_0 \zeta(2) = 1$, the product is one. The factor c_0 in Theorem 1.1 comes only from the relation $\delta_d = c_0 \nu_d$.

1.3 Scope

The result is unconditional. Its ingredients are classical: rank-one interlacing, the min–max principle, smooth-number estimates from [9], and Mertens' theorems. What this paper contributes is the cylinder representation of K, the reduction of its eigenvalue count to a count of cylinder masses, and the evaluation of that count, which together give the exact asymptotic in Theorem 1.1. Section 7 discusses related results on arithmetic matrices, operators on trees, and self-similar spectral problems, and lists comparisons that remain open.

We do not obtain a rate of convergence or a second-order term in (2). The theorem concerns the limiting operators; it does not give a growing-index statement for the finite matrices without an additional estimate (Corollary 6.2). The kernel is positive after a diagonal change of signs, and the proof uses no cancellation in sums of μ; in particular, no new estimate for $M(n)$ follows.

Appendix A gives a second sufficient condition for the transfer from masses to eigenvalues, based on an exact rank-one shift identity, and Appendix B applies it to stationary trees and computes the spectrum of homogeneous trees. Neither appendix is needed for Theorem 1.1. Whether the transfer requires an assumption beyond regular variation of the mass count is left open (Question 7.1).

2 The cylinder representation

The proof separates the geometry of nested sets from an arithmetic counting problem. The relevant diagonal count is

$$D(t) = \#\{d \in \mathcal{S} : \nu_d > t\}. \tag{5}$$

Section 3 gives an abstract criterion for transferring D to the spectrum. Section 4 proves $D(t) \sim [t \log(1/t)]^{-1}$, including the leading constant. In this section we realize K, up to a diagonal unitary and the factor c_0, as a Gram matrix of nested sets whose masses are the numbers ν_d.

2.1 A product probability space

Let $\Omega = \{0,1\}^{\{\text{primes}\}}$, with independent coordinates and probability $1/(p+1)$ that prime p is selected. Denote its probability measure by ν. This is the local model of a random squarefree integer: $1/(p+1)$ is the density, among squarefree integers, of those divisible by p. For $d \in \mathcal{S}$, let

$$A_d = \{\omega : \text{the selected primes at most } P^+(d) \text{ are exactly the prime divisors of } d\}; \qquad A_1 = \Omega.$$

Direct multiplication gives

$$\nu(A_d) = \prod_{p \mid d} \frac{1}{p+1} \prod_{\substack{p \le P^+(d) \\ p \nmid d}} \frac{p}{p+1} = \nu_d. \tag{6}$$

If $j \preceq k$, then $A_k \subseteq A_j$; incomparable indices give disjoint cylinders. Indeed, if $P^+(j) \le P^+(k)$ and the two cylinders intersect, the selected primes at most $P^+(j)$ force $j \preceq k$. Hence

$$\left\langle \mathbf{1}_{A_j}, \mathbf{1}_{A_k} \right\rangle = \nu_{\max(j,k)} \mathbf{1}_{\{j,k \text{ comparable}\}}. \tag{7}$$

The children of d are dq for primes $q > P^+(d)$. Their cylinders cover A_d up to a null set, since $\prod_{p > P^+(d)} p/(p+1) = 0$. Their relative masses are

$$\frac{\nu_{dq}}{\nu_d} = \frac{1}{q+1} \prod_{P^+(d) < p < q} \frac{p}{p+1} \le \frac{1}{P^+(d)+2} \le \frac{1}{3}. \tag{8}$$

For every fixed y there are only finitely many $d \in \mathcal{S}$ with $P^+(d) \le y$. The largest immediate-child ratio therefore tends to zero outside finite sets of vertices, in the sense of Definition 3.3 below. Also $D(t) < \infty$, since $\nu_d \le 1/d$.

2.2 Removing the signs

Let S be the diagonal unitary whose dth entry is $\mu(d)$ on squarefree coordinates and 1 otherwise. On comparable squarefree indices, $\mu(k/j) = \mu(j)\mu(k)$. Equations (1) and (7) therefore identify

$$K = c_0 S V^* V S, \qquad Vx = \sum_{d \in \mathcal{S}} x_d \mathbf{1}_{A_d}, \qquad T = VV^*. \tag{9}$$

Here V is zero on nonsquarefree coordinates; its bounded extension and compactness follow from Proposition 3.1. Since the nonzero spectra of V^*V and VV^* coincide, with multiplicity,

$$\mathsf{n}(K; s) = \mathsf{n}(T; s/c_0). \tag{10}$$

The operator T acts on $L^2(\nu)$ and is the sum over $d \in \mathcal{S}$ of the rank-one operators $f \mapsto \langle f, \mathbf{1}_{A_d} \rangle \mathbf{1}_{A_d}$. The next section studies such sums for general trees of nested sets.

3 Compact cylinder operators and a counting theorem

This section is independent of arithmetic. It shows that a sum of rank-one operators attached to a tree of nested sets is compact, bounds its eigenvalue count above and below by counts of vertices, and gives conditions under which both bounds are asymptotic to the number of sets of mass above the threshold. Section 5 applies it to the tree of Section 2.

Let $\mathcal{T}$ be a countable rooted tree, each vertex having finite depth. On a probability space (Ω, ν), associate to each vertex v a measurable set A_v of positive mass $m_v = \nu(A_v)$, with $A_o = \Omega$ at the root. Child sets lie inside the parent and are pairwise disjoint modulo null sets. They need not cover the parent. Set

$$a_v = \sup_{c \text{ child of } v} \frac{m_c}{m_v}, \qquad a_v = 0 \text{ at a leaf}, \qquad D(t) = \#\{v : m_v > t\}.$$

Throughout this section assume

$$0 \le a_v \le \theta < 1, \qquad 0 < \theta < 1, \qquad D(t) < \infty \quad (t > 0), \qquad M_\theta = \frac{1}{1-\theta}. \tag{11}$$

For f in a Hilbert space, R_f denotes the positive rank-one operator with quadratic form $\langle x, R_f x\rangle = |\langle x, f\rangle|^2$.

Proposition 3.1 (Construction and compactness)**.** *The sum*

$$T = \sum_{v \in \mathcal{T}} R_{\mathbf{1}_{A_v}}$$

defines a positive compact operator on $L^2(\nu)$, with $\|T\| \le M_\theta$. The synthesis map $V : \ell^2(\mathcal{T}) \to L^2(\nu)$ is compact and satisfies $T = VV^*$; the Gram operator V^*V has entries $\nu(A_v \cap A_w)$. If T_v is the sum over descendants of v, including v, then

$$\|T_v\| \le M_\theta m_v.$$

For the finite truncation $T^{(\varepsilon)} = \sum_{m_v > \varepsilon} R_{\mathbf{1}_{A_v}}$, with $0 < \varepsilon < 1$,

$$\left\| T - T^{(\varepsilon)} \right\| \le M_\theta \varepsilon. \tag{12}$$

Proof. At depth ℓ, the sets are disjoint and their masses are at most θ^ℓ. Thus the sum Q_ℓ at that depth is an orthogonal sum of rank-one operators of norms m_v, and $\|Q_\ell\| \le \theta^\ell$. Each Q_ℓ is compact: for every positive threshold only finitely many of these norms exceed it. Hence $\sum_{\ell \ge 0} Q_\ell$ converges in operator norm to a positive compact operator, with norm at most M_θ. The same argument within A_v gives the subtree bound.

The family $\{v : m_v > \varepsilon\}$ is finite and ancestor closed. The omitted vertices are the disjoint subtrees rooted at minimal omitted vertices c, all of mass at most ε. Their operators act on orthogonal spaces $L^2(A_c)$, proving (12).

For any f, the sum of $|\langle f, \mathbf{1}_{A_v}\rangle|^2$ is $\langle f, Tf\rangle$. This defines a bounded analysis operator V^*, whose adjoint is the stated synthesis map. Then $VV^* = T$. Compactness of T implies compactness of V, for example by polar decomposition. The Gram entries follow on coordinate vectors. These constructions also show that finite vertex sums converge in norm: a finite set containing all vertices of mass above ε leaves a positive remainder dominated by the one in (12). □

Use the normalized measure ν/m_v on A_v to define the normalized subtree operator $\widehat{T}_v$. Extension after multiplication by $m_v^{-1/2}$ gives a unitary J_v onto $L^2(A_v, \nu)$, and

$$T_v = m_v J_v \widehat{T}_v J_v^*.$$

Decomposing into child spaces, with the unused part of A_v assigned a zero operator, gives

$$\widehat{T}_v = R_{\mathbf{1}} + \bigoplus_{c \text{ child of } v} \frac{m_c}{m_v} U_c \widehat{T}_c U_c^*. \tag{13}$$

Here U_c is the corresponding normalized embedding and $\|\mathbf{1}\|_{L^2(A_v,\nu/m_v)} = 1$. Consequently

$$1 \le \left\|\widehat{T}_v\right\| \le 1 + M_\theta a_v \le M_\theta. \tag{14}$$

Proposition 3.2 (Tree sandwich)**.** *Let $\mathcal{L}_t$ be the leaves of the finite ancestor-closed tree $\{v : m_v > t\}$. Then*

$$|\mathcal{L}_t| \le \mathsf{n}(T;t) \le \#\{v : m_v \left\|\widehat{T}_v\right\| > t\}. \tag{15}$$

In particular,

$$D(t) - D(t/\theta) \le \mathsf{n}(T;t) \le D((1-\theta)t). \tag{16}$$

Proof. Leaf vertices are incomparable, so the compression of V^*V to their coordinates is diagonal with entries $m_v > t$. The min–max principle and equality of the nonzero spectra of V^*V and VV^* prove the lower bound.

Write $n_v(u) = \mathsf{n}(\widehat{T}_v; u)$. Positive rank-one interlacing in (13) gives

$$n_v(u) \le \mathbf{1}_{\{u<\|\widehat{T}_v\|\}} + \sum_{c \text{ child of } v} n_c(um_v/m_c).$$

For $u \ge \left\|\widehat{T}_v\right\|$ both sides' child counts vanish, which explains the indicator rather than an unrestricted 1. For a fixed $u > 0$, iteration terminates uniformly: at depth h the threshold is at least $u\theta^{-h}$, and all normalized operator norms are at most M_θ. Choose h with $u\theta^{-h} \ge M_\theta$. All contributing vertices have $m_w > um_v/M_\theta$, so only finitely many can contribute. Unrolling proves the upper bound at the root.

A nonleaf of $\{m_v > t\}$ has a child of mass above t, hence has mass above t/θ. This proves the first rough bound. The second follows from (14). □

Definition 3.3. A positive function D is regularly varying at zero with index $-\alpha$ if, for every $b > 0$, $D(bt)/D(t) \to b^{-\alpha}$ as $t \downarrow 0$. We say $a_v \to 0$ outside finite sets if $\{v : a_v > \eta\}$ is finite for every $\eta > 0$.

The second condition is stronger than convergence of a_v to zero along each infinite path of the tree. It requires that, for each $\eta > 0$, all but finitely many vertices have every child of relative mass at most η.

Theorem 3.4 (Transfer from masses to eigenvalues)**.** *Under* (11)*, suppose $a_v \to 0$ outside finite sets and D is regularly varying at zero with index $-\alpha$, where $\alpha > 0$. Then*

$$\mathsf{n}(T;t) \sim D(t) \qquad (t \downarrow 0).$$

The conclusion holds as well for the nonzero spectrum of the Gram operator V^*V.

Proof. Fix $0 < \eta < 1$ and let $F_\eta = \{v : a_v > \eta\}$. Outside F_η, a nonleaf of $\{m_v > t\}$ has mass above t/η, and $\left\|\widehat{T}_v\right\| \le 1 + M_\theta\eta$. Proposition 3.2 gives

$$D(t) - |F_\eta| - D(t/\eta) \le \mathsf{n}(T;t) \le |F_\eta| + D\big(t/(1+M_\theta\eta)\big).$$

Divide by $D(t)$ and let $t \downarrow 0$. Since $\alpha > 0$, $D(t) \to \infty$. The lower limiting ratio is at least $1 - \eta^\alpha$, and the upper limiting ratio is at most $(1 + M_\theta\eta)^\alpha$. Let $\eta \downarrow 0$. □

4 The arithmetic mass count

This section evaluates the count (5). We write each $d > 1$ as its largest prime factor times a squarefree number with smaller prime factors, count the latter with smooth-number estimates, and sum over the largest prime. Because the classical estimates are uniform only in limited ranges, the sum is split into three ranges of the largest prime.

For a prime p, put

$$m(p) = \prod_{q \le p} \frac{q}{q+1}.$$

Let $\Psi(x, y)$ count the positive integers $n \le x$ with $P^+(n) \le y$. Use $\Psi(x, p^-)$ for the strict condition $P^+(n) < p$, and let $\Psi_{\mathrm{sf}}(x; p^-)$ impose squarefreeness as well. All three functions are zero for $x < 1$.

Theorem 4.1. *For the masses ν_d of Section 1,*

$$D(t) = \#\{d \in \mathcal{S} : \nu_d > t\} \sim \frac{1}{t \log(1/t)} \qquad (t \downarrow 0).$$

4.1 Classical inputs and the uniformity actually used

We use Chebyshev's upper bound $\pi(y) \ll y/\log y$ and the two Mertens asymptotics

$$\prod_{p \le y} (1 - 1/p) \sim \frac{e^{-\gamma}}{\log y}, \qquad \sum_{p \le y} \frac{1}{p} = \log\log y + B + o(1).$$

In particular,

$$m(p) \sim \frac{\zeta(2) e^{-\gamma}}{\log p}, \qquad \frac{c}{\log p} \le m(p) \le \frac{C}{\log p} \quad (p \ge 2) \tag{17}$$

for absolute positive constants c, C.

The Dickman function ρ is 1 on $[0, 1]$, zero on $(-\infty, 0)$, and satisfies $u\rho'(u) = -\rho(u-1)$ for $u > 1$. We use:

$$\Psi(x, y) \ll x e^{-u/2} \qquad (x \ge y \ge 2,\ u = \log x/\log y), \tag{18}$$

$$\sup_{u \in I} \left| \frac{\Psi(x, x^{1/u})}{x} - \rho(u) \right| \longrightarrow 0 \qquad (x \to \infty), \tag{19}$$

$$\int_0^\infty \rho(u)\, du = e^\gamma. \tag{20}$$

In (19), I is any fixed compact interval in $(0, \infty)$.

These facts may be sourced to Hildebrand–Tenenbaum [9]. Printed page 414 states the fixed-u Dickman limit. Equation (1.4) there has an extra factor $\log y$ in (18); the following sentence explicitly states that this factor can be removed, citing Tenenbaum's Theorem III.5.1. Thus (18) uses that strengthened bound, not equation (1.4) alone. For (20), use Lemma 2.6 on printed page 426 and the definition of E in (2.3), printed page 422: the Laplace transform is $\exp(\gamma + E(-s))$, and $E(0) = 0$.

For completeness, the compact uniformity in (19) follows from the fixed-u limit. Both $u \mapsto \Psi(x, x^{1/u})/x$ and $\rho(u)$ are nonincreasing; ρ is continuous. A finite mesh on I with small oscillation of ρ brackets all intermediate values between endpoint values, where convergence is simultaneous. For $u < 1$ the smooth count is simply $\lfloor x \rfloor$. No approximation on a growing u-range is used below.

4.2 An exact weak-count identity

Let

$$F(X) = \#\{d \in \mathcal{S} : \nu_d \geq 1/X\} \qquad (X > 1).$$

Every $d > 1$ factors uniquely as $d = pm$ with $p = P^+(d)$ and m squarefree with $P^+(m) < p$. Thus

$$F(X) = 1 + \sum_p \Psi_{\mathrm{sf}}(x_p; p^-), \qquad x_p = \frac{Xm(p)}{p}. \tag{21}$$

Primes $p \geq X$ give zero. We prove $F(X) \sim X/\log X$, then transfer to the strict count. Write $L = \log X$ and $\alpha_p = (\log p)/L$. Fix $0 < a \leq 1/4 < b < 1$ and split the sum at X^a and X^b.

4.3 Small largest primes

The term $p = 2$ equals 1 for all sufficiently large X, since $m = 1$ is the only squarefree positive integer with all prime factors below 2. For $3 \leq p \leq X^a$, (17) implies, uniformly in this range,

$$x_p \geq \frac{cX^{1-a}}{L} \geq p, \qquad \log x_p \geq L/2$$

once X is large. Hence (18) gives

$$\Psi_{\mathrm{sf}}(x_p; p^-) \leq \Psi(x_p, p) \ll \frac{X}{p \log p} \exp\left(-\frac{L}{4 \log p}\right).$$

For integer $k \geq 1$, Chebyshev's bound gives

$$\sum_{e^k < p \leq e^{k+1}} \frac{1}{p} \leq e^{-k}\pi(e^{k+1}) \ll \frac{1}{k}.$$

On such a bin, $1/\log p < 1/k$ and $e^{-L/(4\log p)} \leq e^{-L/(4(k+1))} \leq e^{-L/(8k)}$, since $4(k+1) \leq 8k$ for $k \geq 1$. The small-prime contribution, apart from $p = 2$, is therefore at most

$$CX \sum_{1 \leq k \leq aL} \frac{1}{k^2} e^{-L/(8k)}.$$

The function $f(u) = u^{-2}e^{-L/(8u)}$ is unimodal, with maximum $256e^{-2}/L^2$. Comparing the sum with its integral and twice its maximum, and using

$$\int_0^{aL} u^{-2} e^{-L/(8u)}\, du = \frac{8}{L} e^{-1/(8a)},$$

we obtain

$$\sum_{p \leq X^a} \Psi_{\mathrm{sf}}(x_p; p^-) \ll \frac{X}{L} e^{-1/(8a)} + \frac{X}{L^2} + 1. \tag{22}$$

The implied constant here can be taken independent of a in $(0, 1/4]$.

4.4 The middle range

We first establish, uniformly for $X^a < p < X^b$,

$$\Psi_{\mathrm{sf}}(x_p; p^-) = x_p\{c_0\rho(u_p) + o(1)\}, \qquad u_p = \frac{\log x_p}{\log p}. \tag{23}$$

The squarefree identity $\mu^2(n) = \sum_{\beta^2|n} \mu(\beta)$ gives

$$\Psi_{\mathrm{sf}}(x; p^-) = \sum_{\substack{\beta \le \sqrt{x} \\ P^+(\beta) < p}} \mu(\beta)\Psi(x/\beta^2, p^-).$$

Terms with $\beta > B$ have absolute sum at most $x\sum_{\beta>B}\beta^{-2} \ll x/B$. For fixed B, all $\beta \le B$ satisfy $P^+(\beta) < p$ once X is large. Uniformly in the middle range, $x_p/\beta^2 \to \infty$ and

$$\frac{\log(x_p/\beta^2)}{\log p}$$

stays in a fixed compact subinterval of $(0, \infty)$. Replacing the strict cutoff p^- by p changes a count at argument z by at most z/p. Apply (19), and then continuity of ρ, to obtain

$$\frac{\Psi(x_p/\beta^2, p^-)}{x_p} = \beta^{-2}\rho(u_p) + o(1)$$

uniformly for this finite set of β. First let $X \to \infty$ and then $B \to \infty$. Since $\sum_{\beta\ge1}\mu(\beta)/\beta^2 = 1/\zeta(2) = c_0$, this proves (23).

Equation (17) now gives, uniformly on the middle range,

$$u_p = \frac{1-\alpha_p}{\alpha_p} + o(1), \qquad c_0 x_p = \frac{e^{-\gamma}X}{p\log p}\{1 + o(1)\}.$$

The total mass $\sum_{X^a<p<X^b} x_p$ is $O_{a,b}(X/L)$, so the uniform errors remain $o_{a,b}(X/L)$ after summation. Consequently

$$\frac{L}{X}\sum_{X^a<p<X^b}\Psi_{\mathrm{sf}}(x_p; p^-) = e^{-\gamma}\sum_{X^a<p<X^b}\frac{1}{p}\,\frac{\rho((1-\alpha_p)/\alpha_p)}{\alpha_p} + o(1). \tag{24}$$

Mertens' reciprocal-prime asymptotic implies weak convergence of these prime measures on $[a, b]$ to $d\alpha/\alpha$: on any fixed subinterval $[u, v] \subset [a, b]$ their masses tend to $\log(v/u)$. Approximating the continuous integrand in (24) by step functions yields

$$\lim_{X\to\infty}\frac{L}{X}\sum_{X^a<p<X^b}\Psi_{\mathrm{sf}}(x_p; p^-) = e^{-\gamma}\int_a^b \rho\Big(\frac{1-\alpha}{\alpha}\Big)\,\frac{d\alpha}{\alpha^2} = e^{-\gamma}\int_{(1-b)/b}^{(1-a)/a}\rho(v)\,dv. \tag{25}$$

4.5 Large largest primes and completion

The trivial estimate $\Psi_{\mathrm{sf}}(x_p; p^-) \le x_p$ and (17) show

$$\sum_{X^b\le p<X}\Psi_{\mathrm{sf}}(x_p; p^-) \le \frac{CX}{bL}\sum_{X^b\le p<X}\frac{1}{p} = \frac{CX}{bL}\{\log(1/b) + o(1)\}. \tag{26}$$

All summands in (21) are nonnegative. Thus the limiting middle contribution is a lower bound for $\liminf LF(X)/X$, while (22) and (26) bound the excess in the upper limit. Let $a \downarrow 0$ and $b \uparrow 1$, after taking the limits in X. Equation (20) gives

$$\frac{LF(X)}{X} \longrightarrow e^{-\gamma} \int_0^\infty \rho(v)\, dv = 1.$$

Finally, for every fixed $\varepsilon > 0$,

$$F\big(X/(1+\varepsilon)\big) \le D(1/X) \le F(X).$$

Taking $X \to \infty$ and then $\varepsilon \downarrow 0$ proves Theorem 4.1. In particular, D is regularly varying at zero with index -1.

5 Proof of the main theorem and further consequences

We now combine the cylinder representation, the transfer theorem, and the arithmetic count.

Proof of Theorem 1.1. Equations (8) and (6) verify (11) with $\theta = 1/3$. The child ratios tend to zero outside finite sets. Proposition 3.1 and (9) establish positivity and compactness. Theorems 3.4 and 4.1, followed by (10), give

$$\mathsf{n}(K; s) \sim D(s/c_0) \sim \frac{c_0}{s \log(c_0/s)} \sim \frac{c_0}{s \log(1/s)}.$$

To invert the count without assumptions about multiplicities, put $b_r = c_0/(r \log r)$ for $r \ge 2$. For each fixed $\varepsilon \in (0, 1)$,

$$\frac{\mathsf{n}(K; (1 \pm \varepsilon) b_r)}{r} \longrightarrow \frac{1}{1 \pm \varepsilon}.$$

For all sufficiently large r this traps $\lambda_r(K)$ between $(1-\varepsilon)b_r$ and $(1+\varepsilon)b_r$. Letting $\varepsilon \downarrow 0$ proves (3). Summation, with finitely many initial terms absorbed, gives $\sum_{r \le R} \lambda_r(K) \sim c_0 \log\log R$.

Let $P = R_{\mathbf{1}}$ be the orthogonal projection onto constants in $L^2(\nu)$. The root column of V is $\mathbf{1}$, of norm one, and $Se_1 = e_1$. Thus

$$L = c_0 S V^*(I - P) V S \ge 0.$$

It is compact, and $K - L$ is positive of rank one. Interlacing gives

$$0 \le \mathsf{n}(K; s) - \mathsf{n}(L; s) \le 1.$$

This proves the same counting law for L; inversion and summation apply unchanged. □

Corollary 5.1. *For every $p > 0$, K and L belong to the Schatten class $\mathcal{S}_p$ if and only if $p > 1$. In particular, they are Hilbert–Schmidt but not trace class.*

Proof. For a positive compact operator, membership in $\mathcal{S}_p$ means

$$\sum_r \lambda_r^p < \infty.$$

Compare with $\sum_{r \ge 2} (r \log r)^{-p}$, using Theorem 1.1. □

5.1 A trace-class perturbation, and its precise use

There is also a direct trace-class relation to the diagonal. It is useful for sums of eigenvalues, although the pointwise counting law above uses the tree theorem.

Proposition 5.2. *Let $\Delta = \operatorname{diag}(\delta_d : d \in \mathcal{S})$, with zero entries elsewhere. Then $K - \Delta$ is trace class. If $\delta_r^{\downarrow}$ lists the positive diagonal entries in decreasing order, then, for every $R \geq 1$,*

$$\left| \sum_{r \leq R} \lambda_r(K) - \sum_{r \leq R} \delta_r^{\downarrow} \right| \leq \|K - \Delta\|_{\mathcal{S}_1} .$$

Moreover, for each prime p,

$$\sum_{\substack{d \in \mathcal{S} \\ P^+(d) = p}} \delta_d = \frac{c_0}{p+1}. \tag{27}$$

Proof. For squarefree j, let b_j have coordinate K_{jk} at strict descendants k of j and zero elsewhere. If $j > 1$ and $y = P^+(j)$, write such descendants as $k = jd$. Then $d > y$ and $\delta_{jd}/\delta_j \leq 1/d$, so

$$\|b_j\|^2 \leq \delta_j^2 \sum_{d > y} d^{-2} \leq \frac{\delta_j^2}{y}.$$

The root vector b_1 is in ℓ^2 since $\delta_k \leq c_0/k$. On the other hand,

$$\sum_{P^+(d) = p} \delta_d = \frac{c_0 m(p)}{p} \prod_{q < p} (1 + 1/q) = \frac{c_0}{p+1},$$

proving (27). Hence

$$\sum_{\substack{j \in \mathcal{S} \\ j > 1}} \|b_j\| \leq c_0 \sum_p \frac{1}{(p+1)\sqrt{p}} < \infty.$$

Since $b_j \perp e_j$, the self-adjoint rank-two operator $e_j b_j^* + b_j e_j^*$ has trace norm $2\|b_j\|$. Thus

$$K - \Delta = \sum_{j \in \mathcal{S}} (e_j b_j^* + b_j e_j^*)$$

converges in trace norm; its entries identify it with the stated difference. The eigenvalue-sum bound is Ky Fan's variational principle, using $|\operatorname{Tr}(PE)| \leq \|E\|_{\mathcal{S}_1}$ for finite-rank orthogonal projections P and trace-class E. □

Equation (27) also proves directly that the positive operator K has infinite trace, since $\sum_p 1/(p+1)$ diverges. The trace-class perturbation statement alone is not used to infer (3); the mass scale here is below $1/r$, so a generic trace-class perturbation does not supply the required relative control.

6 Relation to the sparse finite matrices

This section recalls the finite matrices of [12] and the order of limits in their relation to K. Proposition 6.1 is [12, Theorem 4.3]; we include its proof so that the paper can be read on its own. Corollary 6.2 records what an explicit operator-norm estimate would add.

For $1 \le i, j \le n$, let

$$(A_n)_{ij} = \mathbf{1}_{\{i=j\}} + \mathbf{1}_{\{i>1,\ i\in\mathcal{S},\ j=i/P^+(i)\}}, \qquad C_n = A_n^{-1}.$$

The off-diagonal part is strictly lower triangular. Its powers follow successive parent steps, so its finite geometric inverse gives

$$(C_n)_{ij} = \begin{cases} \mu(i/j), & i, j \in \mathcal{S},\ j \preceq i, \\ 1, & i = j \notin \mathcal{S}, \\ 0, & \text{otherwise.} \end{cases} \tag{28}$$

In particular $C_n e_1 = (\mu(1), \dots, \mu(n))^{\mathsf{T}}$.

Proposition 6.1 (Finite Gram convergence)**.** *Extend $H_n = C_n^* C_n/n$ by zero to $\ell^2(\mathbb{N})$. Then*

$$\|H_n - K\|_{\mathrm{HS}} \longrightarrow 0.$$

Consequently, for every fixed $r \ge 1$,

$$\frac{s_r(C_n)}{\sqrt{n}} \longrightarrow \sqrt{\lambda_r(K)},$$

where $s_r(C_n)$ denotes the rth largest singular value.

Proof. For squarefree $j \preceq k \le n$, common descendant rows in (28) give

$$(C_n^* C_n)_{jk} = \mu(k/j)\, G_k(n), \quad G_k(n) = \#\{d \le n/k : \mu^2(d) = 1,\ P^-(d) > P^+(k)\}.$$

Here $P^-(1) = +\infty$. Incomparable squarefree coordinates give zero, and nonsquarefree coordinates give only the diagonal entry 1.

For fixed y, the density of squarefree integers coprime to $P_y = \prod_{p\le y} p$ is

$$\lim_{x\to\infty} \frac{1}{x} \sum_{\substack{d\le x \\ (d,P_y)=1}} \mu^2(d) = \prod_{p\le y}(1-1/p) \prod_{p>y}(1-1/p^2) = c_0 \prod_{p\le y} \frac{p}{p+1}. \tag{29}$$

Indeed, substitute $\mu^2(d) = \sum_{\beta^2|d} \mu(\beta)$. For each fixed β coprime to P_y, the remaining coprimality condition has density $\varphi(P_y)/P_y$; terms $\beta > B$, after division by x, are bounded by $\sum_{\beta>B} \beta^{-2}$. Taking $x \to \infty$ and then $B \to \infty$ gives the Euler product. This proof includes $y = 1$, with $P_y = 1$.

It follows that $G_k(n)/n \to \delta_k$ for each fixed squarefree k. Furthermore $G_k(n)/n \le 1/k$ and $\delta_k \le 1/k$. For maximal coordinate k, there are at most $1 + 2\omega(k)$ ordered comparable pairs. The convergent series

$$\sum_{k\ge1} \frac{1+2\omega(k)}{k^2} < \infty$$

therefore dominates the squared squarefree matrix entries and their differences, up to an absolute constant. Its convergence follows, for example, from $\omega(k) \le \log k/\log 2$ for $k \ge 2$. Dominated convergence gives the Hilbert–Schmidt limit on the squarefree coordinates. The squared Hilbert–Schmidt norm on nonsquarefree coordinates is at most $n/n^2 = 1/n$. Finally, operator-norm convergence and the min–max principle imply convergence of each fixed ordered eigenvalue of H_n. Taking square roots proves the singular-value assertion. □

In particular, the rigorously justified iterated limit is

$$\lim_{r\to\infty}\sqrt{\frac{r\log r}{c_0}}\left(\lim_{n\to\infty}\frac{s_r(C_n)}{\sqrt{n}}\right)=1.$$

A simultaneous limit requires a quantitative error estimate.

Corollary 6.2 (A sufficient condition for growing indices)**.** *Suppose $\varepsilon_n \geq \|H_n - K\|$ and $r_n \leq n$ is an integer sequence with $r_n \to \infty$ and $\varepsilon_n r_n \log r_n \to 0$. Then*

$$s_{r_n}(C_n) \sim \sqrt{\frac{c_0 n}{r_n \log r_n}}.$$

Proof. The min–max principle gives $|\lambda_{r_n}(H_n) - \lambda_{r_n}(K)| \leq \varepsilon_n$. Divide by Theorem 1.1's asymptotic for $\lambda_{r_n}(K)$, then take square roots. □

Remark 6.3 (Bordering and Mertens sums)*.* Let $\mathbf{1}_n$ be the all-ones vector and set

$$B_n = A_n + e_1(\mathbf{1}_n - e_1)^{\mathsf{T}}.$$

Thus the first row is replaced by ones. The determinant lemma, together with (28), gives the exact identity

$$\det B_n = 1 + (\mathbf{1}_n - e_1)^{\mathsf{T}} C_n e_1 = \sum_{j\leq n}\mu(j) = M(n).$$

The present counting theorem is unconditional and concerns the positive limiting Gram spectrum. This determinant identity does not convert it into a new cancellation estimate for $M(n)$. Such an application needs further control of the signed vector and the bordering operation, or uniform finite-size estimates strong enough for the intended arithmetic conclusion.

7 Related work and open questions

7.1 Related work

Arithmetic matrices. The fixed-index part of the connection in Section 6 follows a pattern used by Hilberdink [6] for multiplicative Toeplitz matrices ($f(i/j)\mathbf{1}_{\{j|i\}}$): after normalization, Gram matrices of the finite matrices converge to a positive Hilbert–Schmidt operator, so each fixed singular value has a limit described by an eigenvalue of that operator. Hilberdink and Pushnitski [7, 8] then determined eigenvalue asymptotics for infinite arithmetic matrices built from least common multiples, such as $(n^\sigma m^\sigma/[n,m]^\tau)$, obtaining power laws such as $\lambda_n \sim \kappa n^{-\rho}$. Their proofs use a factorization of these matrices as infinite tensor products over primes, which expresses the eigenvalues as products of local eigenvalues, followed by a Tauberian theorem of Wiener–Ikehara type. They also show that the constant κ need not be the one suggested by the diagonal entries; see (1.5) of [7] and the discussion following it. The kernel K is of a different kind: its entries depend on the ancestor relation in the tree, not on a multiplicative function of the indices, and we have not found a tensor-product factorization of it. Its eigenvalue law also carries a logarithmic factor rather than being a pure power. In contrast with the example of [7], for K the leading constant is the one given by the count of diagonal masses.

Operators on trees. The synthesis map V of (9) is a discrete Hardy-type operator: for finitely supported x, $(Vx)(\omega)$ is the sum of x_d over the vertices d on the path determined by ω. Evans, Harris and Lang [4] proved, for Hardy operators on metric trees with finite vertex degrees and

suitable weights u, v, that the approximation numbers satisfy $n\,a_n \to \alpha_p \int |uv|$, extending earlier Hilbert-space results cited there. When this limit is finite, $a_n = O(1/n)$. For V, Theorem 1.1 and (9) give $a_n(V) = (\lambda_n(K)/c_0)^{1/2} \sim (n \log n)^{-1/2}$, so V is not covered by that theorem with a finite integral. The tree of Section 2 has vertices of infinite degree, but this is a feature of the presentation rather than of the operator. Revealing the primes one at a time gives a binary filtration of Ω whose cylinders B are determined by the selected primes up to some prime p. Each A_d is the cylinder of this kind with $p = P^+(d)$ in which the last revealed prime is selected, and

$$T = \sum_B c_B\, \nu(B)^{-1} R_{\mathbf{1}_B}, \qquad c_B = \begin{cases} \nu(B), & B = \Omega \text{ or the last revealed prime of } B \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

Results on weighted dyadic or martingale Carleson embeddings, and on singular values of weighted operators on trees of bounded degree, are therefore relevant comparisons. The closest results we found are those of Naimark and Solomyak [13] for weighted Hardy-type operators on metric trees of bounded branching. They give two-sided estimates, with unspecified constants, for Schatten and weak Schatten quasi-norms [13, Theorems 4.1 and 4.6, Corollary 4.7], not an eigenvalue asymptotic with a constant. Their weight is an integrable function on the edges, whereas the mass of ν lies on the ends of the tree: the cylinders of the children of a vertex cover its cylinder up to a null set, so no mass is left on the edges. Their theorems therefore do not apply directly. We have not determined whether an approximation argument would yield two-sided bounds for T, and we did not read results on Carleson embeddings at the level of theorems.

Ultrametric spaces. For almost every pair (ω, ω'), the kernel of T is one plus the length of the longest common initial segment of the increasing sequences of primes selected by ω and ω', equivalently, the number of vertices whose cylinders contain both points. This is an ultrametric quantity. Bendikov, Grigor'yan, Pittet and Woess [2] studied isotropic Laplacians on ultrametric spaces with compact balls and a reference measure, which we again call ν; their eigenfunctions are the functions $\mathbf{1}_C/\nu(C) - \mathbf{1}_B/\nu(B)$, for a ball C with parent ball B [2, Theorem 3.8]. The operator T is not diagonal in the analogous basis. For $h = 3\mathbf{1}_{A_2} - 1$, which is this function for the cylinder A_2 and its parent Ω, one has $\langle \mathbf{1}, h \rangle = 0$ and $(Th)(\omega) = 2\sum \nu_d$ for $\omega \in A_2$, the sum taken over the vertices $d \neq 1$ on the path of ω. This is not constant on A_2: on A_6 it is at least $2\nu_2 + 2\nu_6 = 2/3 + 1/6$, while on A_{10} the vertices after 2 have total mass at most $\nu_{10}/(1-1/7) = 7/144 < 1/12$, because every later child ratio is at most $1/7$ by (8); so $Th < 2/3+1/6$ there. Hierarchical Schrödinger operators, which add a potential to a hierarchical Laplacian, are studied by Bendikov, Grigor'yan and Molchanov [1]. They assume a homogeneous ultrametric space that is not compact [1, Section 2.1], and their spectral results concern self-adjointness, discreteness of the spectrum when the potential tends to infinity, and the number of negative eigenvalues [1, Theorem 2.1]. They do not address eigenvalue asymptotics for a compact positive operator such as T, whose underlying space is compact and whose masses are not homogeneous.

Self-similar spectral problems. The recursion behind Proposition 3.2, in which a subtree operator is a rank-one operator plus rescaled copies of smaller subtree operators, resembles the recursive counting identities used for spectral asymptotics on self-similar sets, where renewal theory supplies the asymptotics; see Kigami and Lapidus [10]. Appendix B uses the key renewal theorem in exactly this way for stationary trees. For the arithmetic tree the subtrees are not rescaled copies of a single operator, and the asymptotics come instead from comparison with vertex counts.

The arithmetic count. Theorem 4.1 is an application of classical estimates for smooth numbers [9], of the same kind as those used for sums involving the largest prime factor; for instance, Erdős, Ivić and Pomerance [3] evaluate $\sum_{n \le x} 1/P^+(n)$ through $\Psi(x, y)$ and ρ. We do not regard the count as new in method.

We have not found Theorem 1.1 in the sources discussed above. The comparisons left incomplete above, and the full text of Kigami–Lapidus [10], which we could not obtain, could change that assessment.

7.2 Open questions

Question 7.1. Under (11), suppose D is regularly varying at zero with index -1. Must $\mathsf{n}(T;t) \sim D(t)$ hold without either vanishing child ratios or the profile hypotheses of Theorem A.4?

The exact decomposition (32) reduces this question to the locations of unit-integral, zero–one profiles at varying mass scales. Their common integral does not, by itself, justify exchanging a limit and the vertex sum. Proposition A.3 gives an integrated comparison without profile hypotheses, but does not establish a pointwise counting asymptotic. Proposition B.2 is not a counterexample to Question 7.1, because it lacks the stipulated regular variation.

Trees whose splitting rule depends on the position of a vertex's mass on a logarithmic scale are natural candidates for a negative answer. For such trees we have proved neither that the mass count is regularly varying nor that $\mathsf{n}(T;t)/D(t)$ fails to converge, and we make no claim about them here.

Question 7.2. Can the proof of Theorem 4.1 and the sandwich in Proposition 3.2 be made quantitative enough to give a useful remainder in (2)?

The proof as written takes $X \to \infty$ for fixed a, b, then sends $a \downarrow 0$ and $b \uparrow 1$. The compact-u input and this order of limits yield a leading equivalent only. In particular, the present argument does not identify a second term or a rate.

Question 7.3. Can one bound $\|C_n^* C_n/n - K\|$ explicitly in a range that makes Corollary 6.2 effective for useful r_n?

Hilbert–Schmidt convergence suffices for fixed indices, but supplies no explicit scale in the proof given here. A quantitative treatment must control both the rough-number densities and the matrix tail, then compare that error with $1/(r_n \log r_n)$.

Remark 7.4 (Arithmetic reach). The main result concerns a positive kernel whose signs can be removed by a diagonal unitary. Its proof uses classical smooth-number estimates and does not produce new cancellation in Möbius sums. An application to a zero-free region would require an additional theorem that controls the relevant signed arithmetic quantity; the asymptotic $\lambda_r(K) \sim c_0/(r \log r)$ is not such a theorem by itself.

Remark 7.5 (Attribution). The compactness argument, interlacing, the rank-one trace identity, the renewal theorem, and the smooth-number inputs belong to established operator theory, probability, and analytic number theory. We make no priority claim for the abstract tree results in Section 3 or in the appendices.

Role of AI

AI assisted the research development, literature search, proof exposition, and mathematical checking under the author's direction. Process-separated AI reviews are not independent expert peer review, and no independent human expert has examined the proofs. The release repository records the inspected sources, the proof reviews, and their dispositions.

A Spectral shifts and a weaker transfer hypothesis

This appendix gives an alternative sufficient condition for transfer. It is not needed for Theorem 1.1. The tree assumptions are (11) throughout.

A.1 Unit-integral rank-one shifts

Lemma A.1. *Let B be positive and compact on a Hilbert space, and let $P = R_\phi$. For $t > 0$, put*

$$\xi(t) = \mathsf{n}(B + P; t) - \mathsf{n}(B; t).$$

Then $\xi(t) \in \{0, 1\}$, it vanishes for $t \geq \|B + P\|$, and

$$\int_0^\infty \xi(t)\, dt = \|\phi\|^2 . \tag{30}$$

No trace-class assumption on B is required.

Proof. Positivity and rank-one interlacing give $0 \leq \xi \leq 1$. The support bound is immediate. For $s > 0$, both positive parts below have finite rank and the layer-cake formula gives

$$g(s) := \operatorname{Tr}(B + P - sI)_+ - \operatorname{Tr}(B - sI)_+ = \int_s^\infty \xi(t)\, dt.$$

If F is the spectral projection of $B + P$ on (s, ∞), the variational principle for the trace of the positive part gives

$$g(s) \leq \operatorname{Tr}(FP) \leq \|\phi\|^2 .$$

For the reverse bound, let $E = \mathbf{1}_{(s,\infty)}(B)$ and $\psi = (I - E)\phi$. If $\psi = 0$, the trial projection E gives $g(s) \geq \|\phi\|^2$. Otherwise take the trial projection $Q = E + R_{\psi/\|\psi\|}$. Since $Q\phi = \phi$,

$$g(s) \geq \|\phi\|^2 + \langle \psi/\|\psi\|, B(\psi/\|\psi\|)\rangle - s \geq \|\phi\|^2 - s.$$

Let $s \downarrow 0$ and use monotone convergence of the nonnegative integral. □

For each vertex set

$$\widehat{B}_v = \widehat{T}_v - R_{\mathbf{1}}, \qquad \widehat{\xi}_v(u) = \mathsf{n}(\widehat{T}_v; u) - \mathsf{n}(\widehat{B}_v; u).$$

Then

$$\widehat{\xi}_v \in \{0, 1\}, \qquad \widehat{\xi}_v(u) = 0 \ (u \geq M_\theta), \qquad \int_0^\infty \widehat{\xi}_v(u)\, du = 1. \tag{31}$$

Proposition A.2 (Exact counting decomposition)**.** *For every $t > 0$,*

$$\mathsf{n}(T; t) = \sum_v \widehat{\xi}_v(t/m_v), \qquad D(t) = \sum_v \mathbf{1}_{\{t/m_v < 1\}}. \tag{32}$$

Only finitely many terms in either sum can be nonzero.

Proof. The decomposition (13) gives

$$\mathsf{n}(T_v; t) = \widehat{\xi}_v(t/m_v) + \sum_{c \text{ child of } v} \mathsf{n}(T_c; t).$$

Unroll this identity from the root. At depth h, every subtree norm is at most $M_\theta \theta^h$, so all residual counts vanish once this is at most t. Nonzero shifts require $m_v > t/M_\theta$, a finite set. The identity for D is its definition. □

Proposition A.3 (An integrated comparison)**.** *For every $t > 0$,*

$$0 \le \int_t^\infty \{\mathsf{n}(T;s) - D(s)\}\, ds \le tD(t/M_\theta). \tag{33}$$

Proof. For each vertex the function $f_v(s) = \widehat{\xi}_v(s/m_v)$ takes values in $[0,1]$, is supported in $[0, M_\theta m_v]$, and has integral m_v. If $t < m_v$, then

$$\int_t^\infty (f_v(s) - \mathbf{1}_{\{s<m_v\}})\, ds = t - \int_0^t f_v(s)\, ds \in [0,t].$$

If $t \ge m_v$, the same expression is $\int_t^\infty f_v(s)\, ds$, which lies in $[0, m_v] \subseteq [0,t]$. It is zero when $t \ge M_\theta m_v$. Sum over the finite set of possible contributors using (32). □

A.2 Transfer through limiting profiles

The following precise domination hypothesis is stronger than pointwise convergence of the profiles. It states what is needed to exchange the varying vertex sum with a limiting profile.

Theorem A.4 (Profile transfer at index -1)**.** *Suppose D is regularly varying at zero with index -1. Partition the vertices into an exceptional class $\mathcal{E}$ and finitely many classes $\mathcal{V}_1, \dots, \mathcal{V}_q$. Define*

$$D_{\mathcal{U}}(t) = \#\{v \in \mathcal{U} : m_v > t\}.$$

Assume

$$D_{\mathcal{E}}(t) = o(D(t)), \qquad \frac{D_{\mathcal{V}_k}(t)}{D(t)} \longrightarrow \beta_k > 0, \qquad \sum_{k=1}^q \beta_k = 1.$$

For each k, suppose there is a bounded Riemann-integrable function h_k on $[0, M_\theta]$, extended by zero outside that interval. For every $\varepsilon > 0$, suppose there is a nonnegative bounded Riemann-integrable function $F_{k,\varepsilon}$ on the same interval, also extended by zero, such that

$$\int_0^{M_\theta} F_{k,\varepsilon}(u)\, du \longrightarrow 0 \quad (\varepsilon \downarrow 0),$$

and, outside a finite subset of $\mathcal{V}_k$ depending on ε,

$$|\widehat{\xi}_v(u) - h_k(u)| \le F_{k,\varepsilon}(u) \qquad \text{for every } u > 0. \tag{34}$$

Then $\mathsf{n}(T;t) \sim D(t)$.

Proof. We first record a weighted-count consequence of regular variation. If $\mathcal{U} = \mathcal{V}_k$ and g is bounded and Riemann integrable on $[0, M_\theta]$, extended by zero, then

$$\frac{1}{D(t)} \sum_{v \in \mathcal{U}} g(t/m_v) \longrightarrow \beta_k \int_0^{M_\theta} g(u)\, du. \tag{35}$$

To see this, the normalized counting measure of the points t/m_v assigns mass

$$\frac{D_{\mathcal{U}}(t/b)}{D(t)} \longrightarrow \beta_k b$$

to $(0, b)$, for each $b > 0$. Its mass at a fixed positive endpoint tends to zero: bracket that mass by the difference between the counts at $b - \delta$ and $b + \delta$, take the limit, and let $\delta \downarrow 0$. Mass near zero is bounded by the mass of $(0, \delta)$, whose limit is $\beta_k \delta$. Step-function upper and lower bounds, or Darboux sums on a fixed finite partition, now prove (35). This also accounts for possible atoms at M_θ.

Each class $\mathcal{V}_k$ is infinite. Integrating (34) at any vertex outside its finite exception and using (31) gives

$$\left| 1 - \int h_k \right| \le \int F_{k,\varepsilon}.$$

Thus $\int h_k = 1$. For fixed ε, the finitely many exceptional vertices in each class contribute $O(1)$ after subtracting the corresponding h_k terms. Apply (35) to both h_k and $F_{k,\varepsilon}$ in (34). The upper limiting normalized error is at most $\beta_k \int F_{k,\varepsilon}$.

Finally the entire class $\mathcal{E}$ contributes at most $D_{\mathcal{E}}(t/M_\theta) = o(D(t))$, by regular variation at the fixed scale $1/M_\theta$. Sum over k and let $\varepsilon \downarrow 0$. Equation (32) then gives the limit $\sum_k \beta_k = 1$. □

Remark A.5 (Recovery of vanishing child ratios). When $a_v \le \eta$, (13) gives

$$\left\| \widehat{B}_v \right\| \le M_\theta \eta, \qquad 1 \le \lambda_1(\widehat{T}_v) \le 1 + M_\theta \eta, \qquad \lambda_2(\widehat{T}_v) \le M_\theta \eta.$$

Consequently $\widehat{\xi}_v(u) = 1$ for $M_\theta \eta < u < 1$, and it is zero for $u \ge 1 + M_\theta \eta$. Thus the common profile $h = \mathbf{1}_{[0,1)}$ works in Theorem A.4, with error envelope

$$F_\eta = \mathbf{1}_{[0, M_\theta \eta]} + \mathbf{1}_{[1, 1 + M_\theta \eta]}$$

restricted to $[0, M_\theta]$, for sufficiently small η. Its integral is at most $2 M_\theta \eta$. This recovers the index--1 case of Theorem 3.4. The argument uses only the upper bound $\left\| \widehat{T}_v \right\| - 1 \le \left\| \widehat{B}_v \right\| \le M_\theta \eta$; no lower bound on $\left\| \widehat{T}_v \right\| - 1$ is needed.

Remark A.6. At index $-\alpha$, the measure in (35) would instead be $\beta_k \alpha u^{\alpha - 1}\, du$. The unit-integral identity (31) fixes its contribution without additional moments precisely when $\alpha = 1$. The profile theorem is therefore stated only at that index.

B Stationary trees and a lattice comparison

Proposition B.1 (Finite stationary non-arithmetic trees)**.** *Fix $k \ge 2$ and $\rho_1, \dots, \rho_k > 0$ with $\sum_i \rho_i = 1$. On $\{1, \dots, k\}^{\mathbb{N}}$ with the corresponding product probability, take the cylinders indexed by all finite words, including the empty word. Let T be their cylinder operator. Put*

$$a_i = -\log \rho_i, \qquad H = \sum_{i=1}^{k} \rho_i a_i.$$

Assume there is no $h > 0$ such that every a_i belongs to $h\mathbb{Z}$. Then, as $t \downarrow 0$,

$$D(t) \sim \mathsf{n}(T; t) \sim \frac{1}{Ht}.$$

Proof. The masses are $m_w = \prod_{j=1}^{|w|} \rho_{w_j}$, and $\theta = \max_i \rho_i < 1$. Vertices with mass above t have bounded depth, so $D(t) < \infty$. Proposition 3.1 applies.

Let F be the probability measure assigning mass ρ_i to a_i, combining equal atoms if necessary, and let $U = \sum_{\ell \geq 0} F^{*\ell}$ be its renewal measure, including the unit atom at zero. For $g(x) = e^{-x}\mathbf{1}_{\{x>0\}}$ and $z(x) = e^{-x}D(e^{-x})$, direct summation over words gives

$$z(x) = \int_{[0,\infty)} g(x-y)\, dU(y).$$

Indeed, a word contributes probability $m_w = e^{-\sum_j a_{w_j}}$ to the renewal measure, cancelling the factor $e^{\sum_j a_{w_j}}$ in $g(x - \sum_j a_{w_j})$.

The increments have finite positive mean H and are non-arithmetic by hypothesis. The function g is directly Riemann integrable. The key renewal theorem therefore gives

$$z(x) \longrightarrow \frac{1}{H}\int_0^\infty e^{-u}\, du = \frac{1}{H}.$$

For a precise source, see Gallager [5], Theorem 4.7.3, printed page 202. That theorem uses the renewal measure without the initial atom; the additional term here is $g(x) \to 0$. Thus $D(t) \sim 1/(Ht)$.

All normalized subtrees are unitarily equivalent. Their shifts $\widehat{\xi}_v$ are consequently one common function h. It is bounded by one and supported in $[0, M_\theta]$, and has integral one by Lemma A.1. Away from zero its discontinuities are among finitely many eigenvalues on each compact subinterval; it is therefore Riemann integrable on $[0, M_\theta]$. Apply Theorem A.4 with one class and zero error envelopes to get $\mathsf{n}(T;t) \sim D(t)$. □

For example, the weights $(3/10, 7/10)$ satisfy the non-arithmetic condition. A rational relation between $\log(10/3)$ and $\log(10/7)$ would, after exponentiation and comparison of prime exponents, give a nontrivial equality between powers involving distinct primes.

Proposition B.2 (Homogeneous k-ary trees)**.** *In the same construction with $\rho_i = 1/k$ for all i, the spectrum is*

$$\frac{k}{k-1} \quad \textit{with multiplicity } 1, \qquad \frac{k^{-\ell}}{k-1} \quad \textit{with multiplicity } (k-1)k^\ell \quad (\ell \geq 0).$$

In particular,

$$\mathsf{n}(T;t) = \mathbf{1}_{\{t<k/(k-1)\}} + (k-1)D((k-1)t), \qquad D(t) = \sum_{\ell \geq 0} k^\ell \mathbf{1}_{\{k^{-\ell}>t\}}. \tag{36}$$

The function D is not regularly varying at zero with index -1. For $k = 3$,

$$\liminf_{t\downarrow 0} \frac{\mathsf{n}(T;t)}{D(t)} = \frac{2}{3}, \qquad \limsup_{t\downarrow 0} \frac{\mathsf{n}(T;t)}{D(t)} = 2.$$

Proof. Let E_ℓ be conditional expectation onto the first ℓ coordinates, with E_0 the projection onto constants. The sum of cylinder rank-one operators at depth ℓ is $k^{-\ell}E_\ell$. Thus $T = \sum_{\ell \geq 0} k^{-\ell}E_\ell$. Constants have eigenvalue $\sum_{\ell \geq 0} k^{-\ell} = k/(k-1)$. The martingale difference space $\operatorname{ran}(E_{\ell+1} - E_\ell)$ has dimension $(k-1)k^\ell$, and T acts on it by $\sum_{j \geq \ell+1} k^{-j} = k^{-\ell}/(k-1)$. These orthogonal spaces and the constants exhaust L^2. This proves the spectrum and (36).

For any fixed $c \in (1, k)$ and $t_m = k^{-m}$, the strict counting convention gives $D(ct_m) = D(t_m)$. This is inconsistent with the ratio c^{-1} required for regular variation of index -1.

For $k = 3$, in each mass interval the argument $2t$ crosses exactly one mass level. The ratios $D(2t)/D(t)$ have limiting values $1/3$ and 1, by summing the geometric series in (36). The single constant eigenvalue is negligible as $t \downarrow 0$, giving the two stated limits. □

Remark B.3. For $k = 2$, (36) instead gives $\mathsf{n}(T; t) = D(t) + 1$ for $0 < t < 2$. Thus a nonvanishing child ratio can coexist with exact asymptotic transfer. The $k = 3$ example shows that $\theta < 1$ alone does not force transfer, but it does not answer the question with regular variation retained: its D fails that hypothesis.

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